

Continuous Dynamics

smooth and non smooth motion

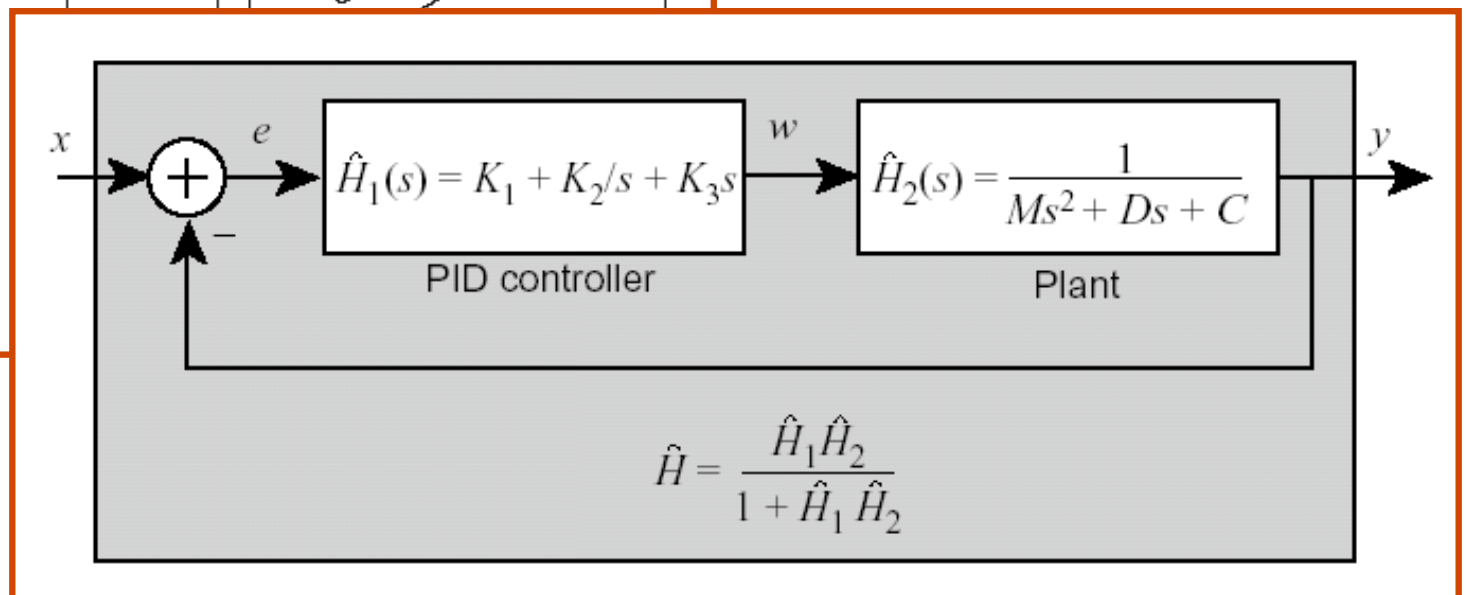
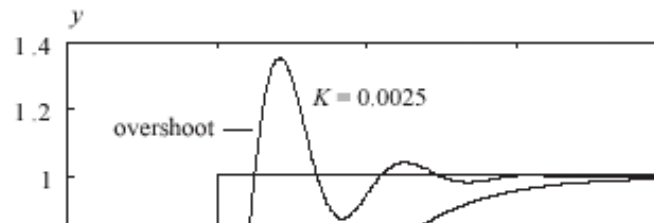
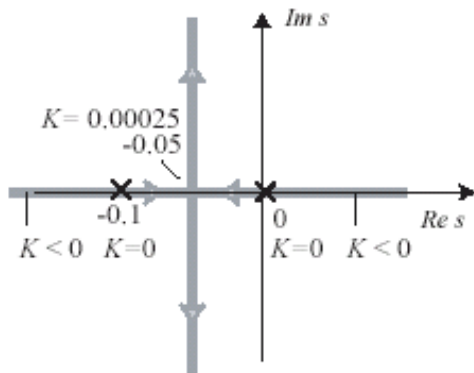
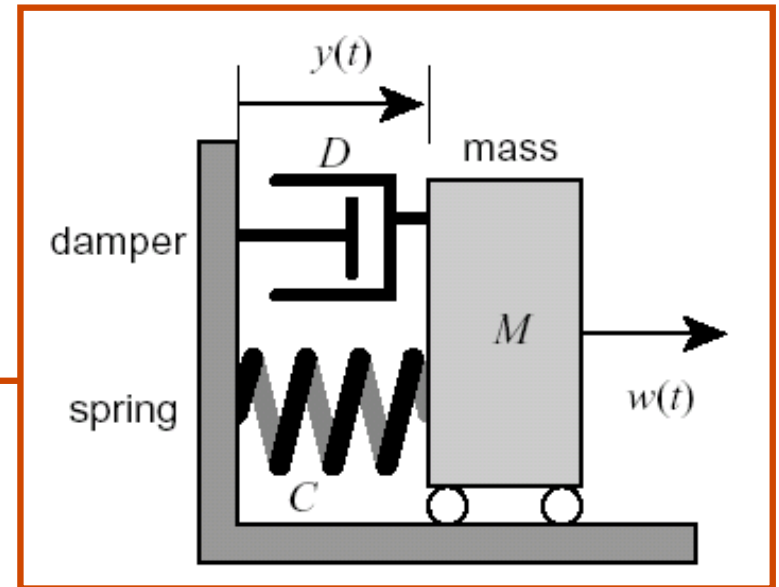
- smooth: notion of linearity, time invariance, and continuity
- non smooth: abrupt, discrete, one smooth motion translates into other smooth motion
- joint smooth and non smooth motion — hybrid systems
- combination of discrete and continuous behaviors is close to modeling of cyber and physical system.

Modeling a Dynamical System

- Modeling Method: ordinary differential equations(ODEs).....
- Modeling Language: LabVIEW, Simulink(from matlab)
- Properties of such models: linearity, time invariance, stability,

Modeling of Continuous Dynamics

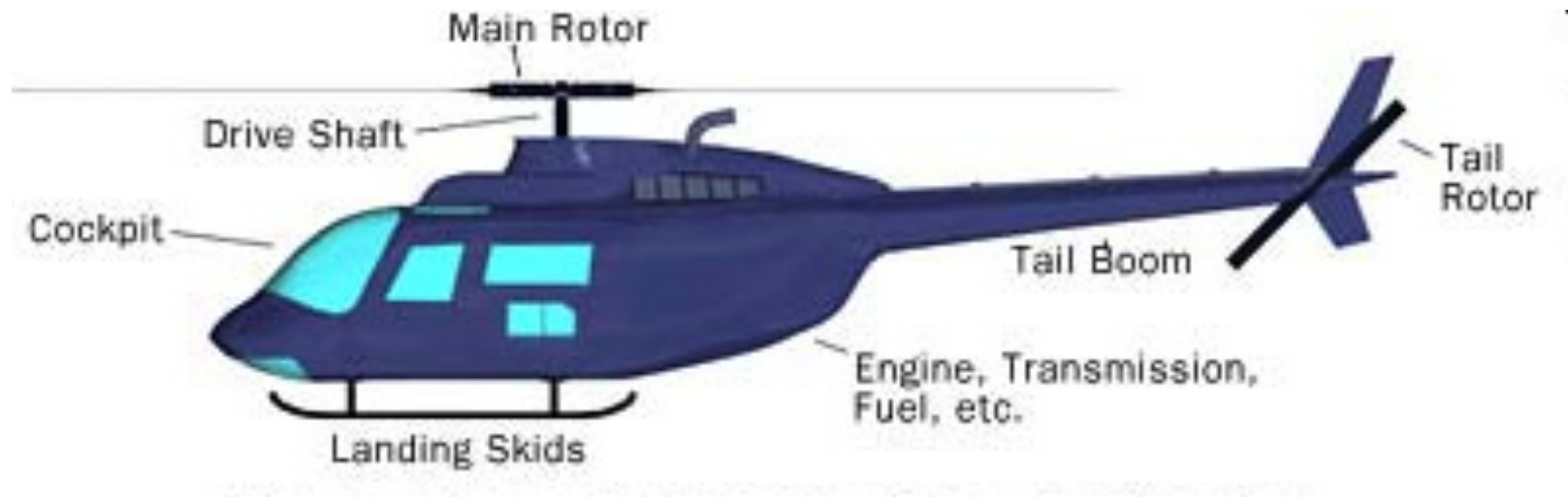
Ordinary differential equations, Laplace transforms, feedback control systems, stability analysis, robustness analysis, ...



Continuous Dynamics

- Classical mechanics
- smooth motion
 - ✦ linearity
 - ✦ time invariance
 - ✦ continuity
- modeling with tool
 - ✦ matlab

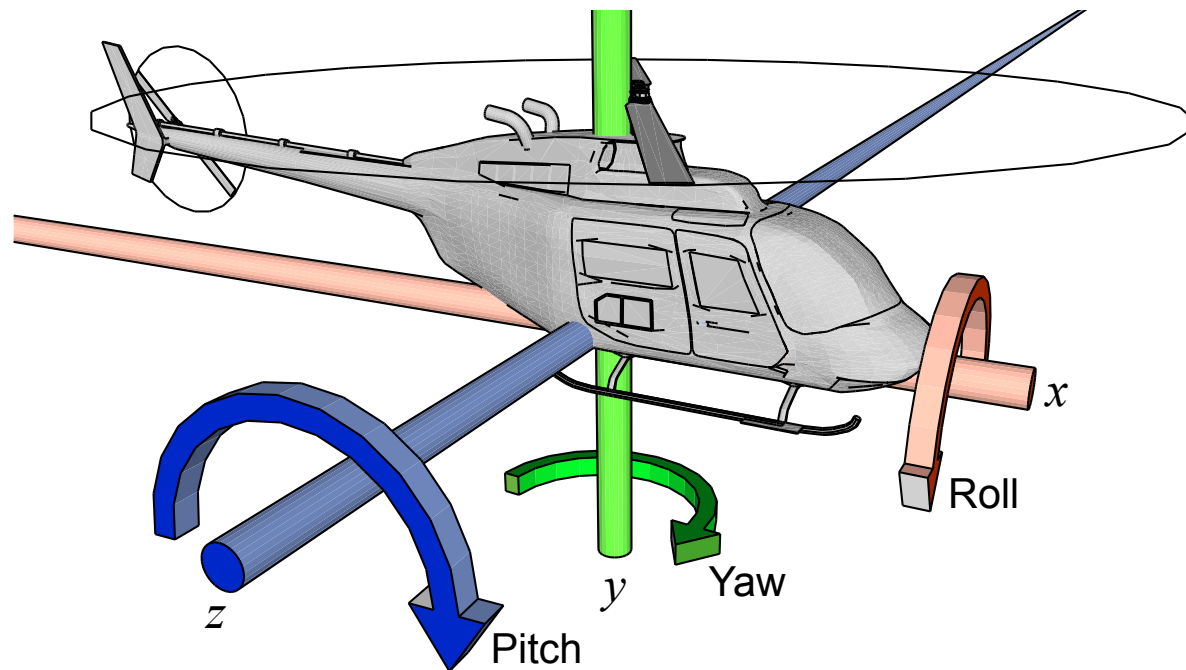
An Example: Modeling Helicopter Dynamics



Modeling Physical Motion

Six degrees of freedom:

- Position: x , y , z
- Orientation: roll (橫滾), yaw (偏航), pitch(俯仰)



Notation

- Position is given by three functions:

$$x: \mathbb{R} \rightarrow \mathbb{R}$$

$$y: \mathbb{R} \rightarrow \mathbb{R}$$

$$z: \mathbb{R} \rightarrow \mathbb{R}$$

- where the domain represents time and
- the codomain represents either distance along an axis.
- Collecting into a vector:

$$\mathbf{x}: \mathbb{R} \rightarrow \mathbb{R}^3$$

Position at time $t \in \mathbb{R}$ is $\mathbf{x}(t) \in \mathbb{R}^3$

Notation

Velocity

$$\dot{\mathbf{x}}: \mathbb{R} \rightarrow \mathbb{R}^3$$

is the derivative, $\forall t \in \mathbb{R}$,

$$\dot{\mathbf{x}}(t) = \frac{d}{dt}\mathbf{x}(t)$$

Acceleration $\ddot{\mathbf{x}}: \mathbb{R} \rightarrow \mathbb{R}^3$ is the second derivative,

$$\ddot{\mathbf{x}} = \frac{d^2}{dt^2}\mathbf{x}$$

Force on an object is $\mathbf{F}: \mathbb{R} \rightarrow \mathbb{R}^3$.

Newton's Second Law

Newton's second law

$$\mathbf{F}(t) = M\ddot{\mathbf{x}}(t)$$

M is the mass of the object

$$\forall t > 0, \quad \dot{\mathbf{x}}(t) = \dot{\mathbf{x}}(0) + \int_0^t \ddot{\mathbf{x}}(\tau) d\tau$$

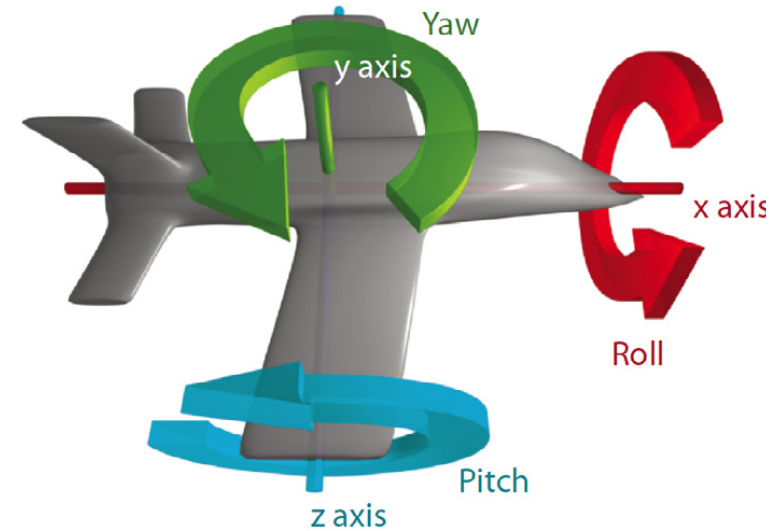
$$\forall t > 0, \quad \dot{\mathbf{x}}(t) = \dot{\mathbf{x}}(0) + \frac{1}{M} \int_0^t \mathbf{F}(\tau) d\tau,$$

Newton's Second Law

$$\begin{aligned}\mathbf{x}(t) &= \mathbf{x}(0) + \int_0^t \dot{\mathbf{x}}(\tau) d\tau \\ &= \mathbf{x}(0) + t\dot{\mathbf{x}}(0) + \frac{1}{M} \int_0^t \int_0^\tau \mathbf{F}(\alpha) d\alpha d\tau,\end{aligned}$$

Orientation

- Orientation: $\theta: \mathbb{R} \rightarrow \mathbb{R}^3$
- Angular velocity: $\dot{\theta}: \mathbb{R} \rightarrow \mathbb{R}^3$
- Angular acceleration: $\ddot{\theta}: \mathbb{R} \rightarrow \mathbb{R}^3$
- Torque: $\mathbf{T}: \mathbb{R} \rightarrow \mathbb{R}^3$



$$\dot{\theta}(t) = \begin{bmatrix} \dot{\theta}_x(t) \\ \dot{\theta}_y(t) \\ \dot{\theta}_z(t) \end{bmatrix} = \begin{bmatrix} \text{roll} \\ \text{yaw} \\ \text{pitch} \end{bmatrix}$$

angular acceleration

$$\mathbf{T}(t) = \frac{d}{dt} (\mathbf{I}(t) \dot{\boldsymbol{\theta}}(t))$$

- $\mathbf{T}$ is the torque (扭力) vector in three axes.
- $\mathbf{I}(t)$ is the moment of inertia tensor(惯性张量) of the object. a 3×3 matrix that depends on the geometry (几何形状) and orientation (方位) of the object.

$$\begin{bmatrix} T_x(t) \\ T_y(t) \\ T_z(t) \end{bmatrix} = \frac{d}{dt} \left(\begin{bmatrix} I_{xx}(t) & I_{xy}(t) & I_{xz}(t) \\ I_{yx}(t) & I_{yy}(t) & I_{yz}(t) \\ I_{zx}(t) & I_{zy}(t) & I_{zz}(t) \end{bmatrix} \begin{bmatrix} \dot{\theta}_x(t) \\ \dot{\theta}_y(t) \\ \dot{\theta}_z(t) \end{bmatrix} \right)$$

simple example

$$\mathbf{T}(t) = I\ddot{\theta}(t), \quad \text{where a constant scalar } I \text{ for a spherical}$$

Rotational velocity is the integral of acceleration

$$\dot{\theta}(t) = \dot{\theta}(0) + \int_0^t \ddot{\theta}(\tau) d\tau, \quad \dot{\theta}(t) = \dot{\theta}(0) + \frac{1}{I} \int_0^t \mathbf{T}(\tau) d\tau.$$

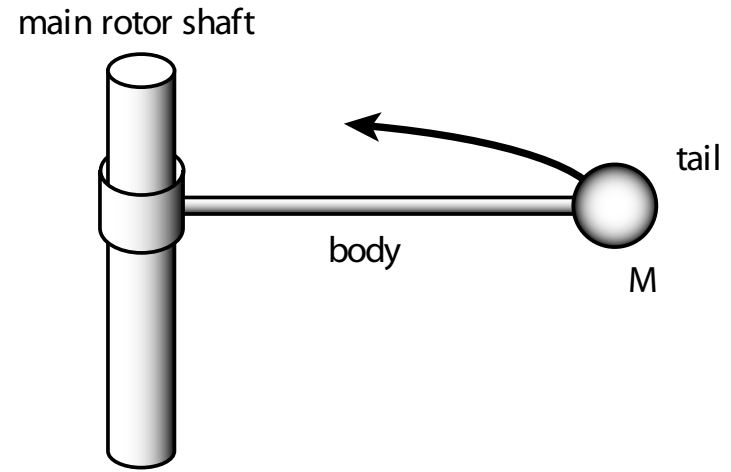
$$\begin{aligned} \theta(t) &= \theta(0) + \int_0^t \dot{\theta}(\tau) d\tau \\ &= \theta(0) + t\dot{\theta}(0) + \frac{1}{I} \int_0^t \int_0^\tau \mathbf{T}(\alpha) d\alpha d\tau \end{aligned}$$

example

- model-order reduction: reduce the number of dimension that are consider
- a moving vehicle on a flat surface
- orientation only consider y axis

example

- A helicopter with two rotors
 - ✦ position is fixed.
 - ✦ the tail rotor counteracts the main rotor's spinning
 - ✦ remains vertical, pitch and roll are zero, only has yaw
- consider the tail rotor

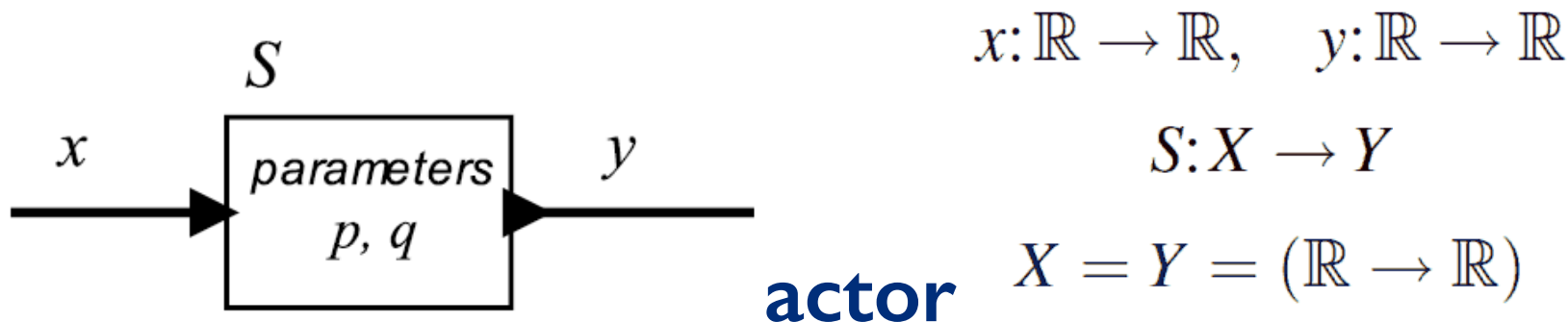


$$\ddot{\Theta}_y(t) = T_y(t)/I_{yy}.$$

$$\dot{\Theta}_y(t) = \dot{\Theta}_y(0) + \frac{1}{I_{yy}} \int_0^t T_y(\tau) d\tau.$$

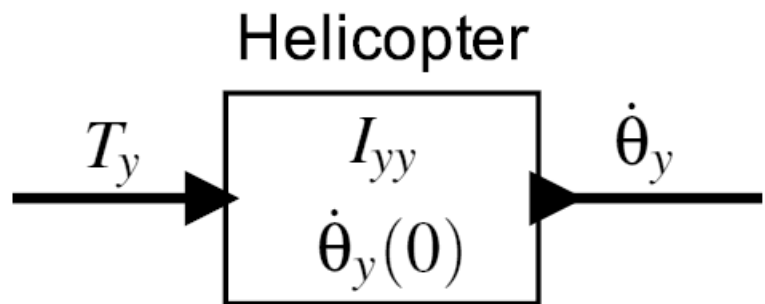
Actor Model of Systems

- A *system* is a function that accepts an input *signal* and yields an output signal.
- In a large system, it is a *component*
- The domain and range of the system function are sets of signals, which themselves are functions.
- *Parameters* may affect the definition of the function S .



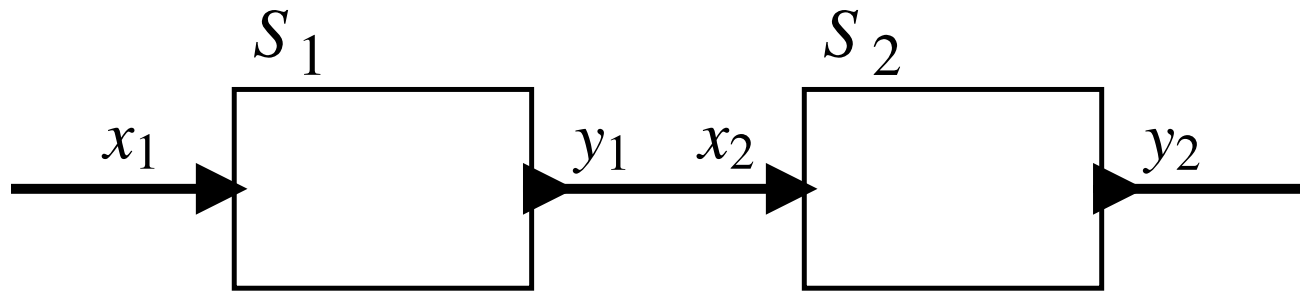
Actor model of the helicopter

- Input is the net torque of the tail rotor and the top rotor.
- Output is the angular velocity around the y axis.
- Parameters of the model are shown in the box.
- The input and output relation is given by the equation to the right.



$$\dot{\theta}_y(t) = \dot{\theta}_y(0) + \frac{1}{I_{yy}} \int_0^t T_y(\tau) d\tau$$

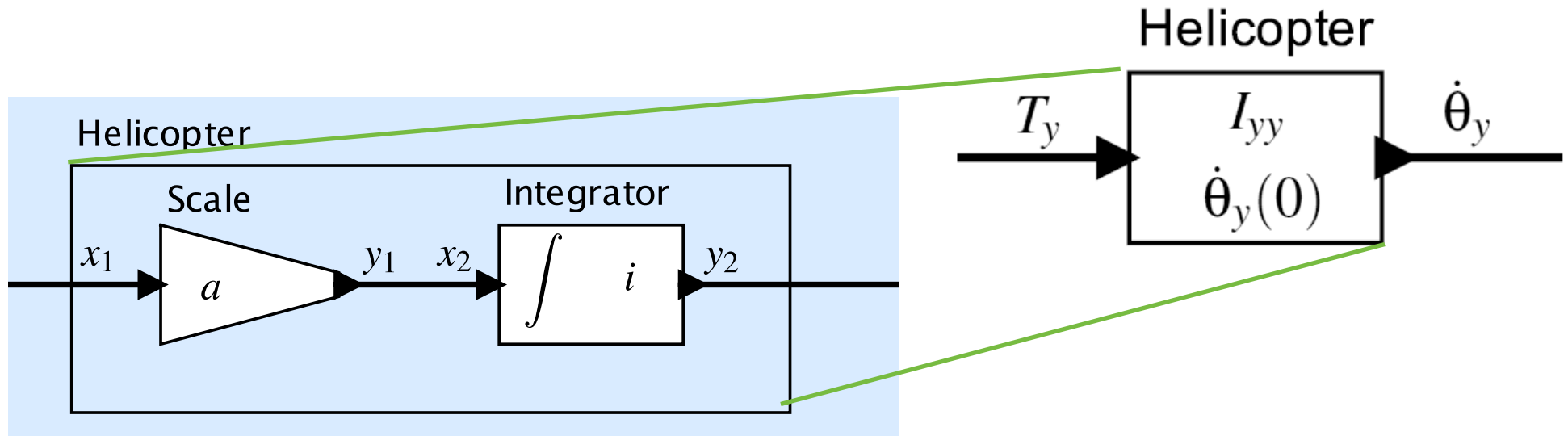
Actor models is composable



- Cascade composition: the “wire” between the output of S_1 and the input of S_2 means precisely that $y_1 = x_2$,

$$\forall t \in \mathbb{R}, y_1(t) = x_2(t).$$

helicopter example

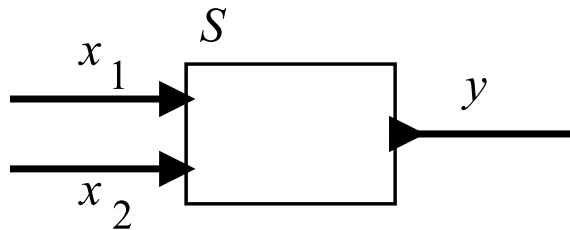


$$\forall t \in \mathbb{R}, \quad y_1(t) = ax_1(t).$$

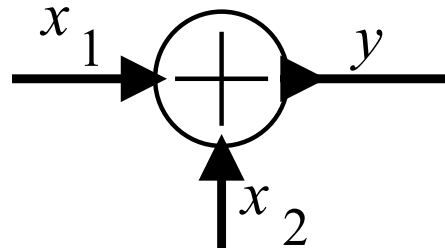
$$\forall t \in \mathbb{R}, \quad y_2(t) = i + \int_0^t x_2(\tau) d\tau.$$

$$a = 1/I_{yy} \text{ and } i = \dot{\theta}_y(0)$$

Actor models with multiple inputs

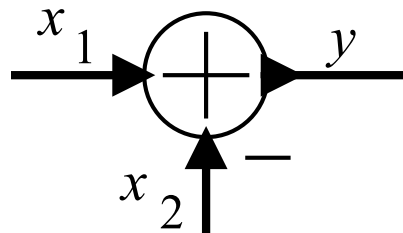


- two input signals and one output signal



- adder

$$\forall t \in \mathbb{R}, y(t) = x_1(t) + x_2(t).$$



- one of the inputs will be subtracted rather than added

$$S : (\mathbb{R} \rightarrow \mathbb{R})^2 \rightarrow (\mathbb{R} \rightarrow \mathbb{R})$$

$$\forall t \in \mathbb{R}, \forall x_1, x_2 \in (\mathbb{R} \rightarrow \mathbb{R}),$$

$$(S(x_1, x_2))(t) = y(t) = x_1(t) - x_2(t).$$